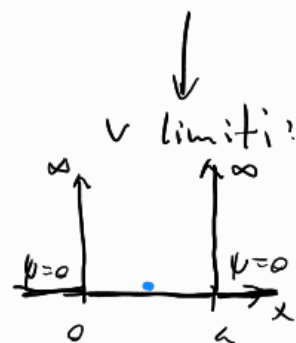
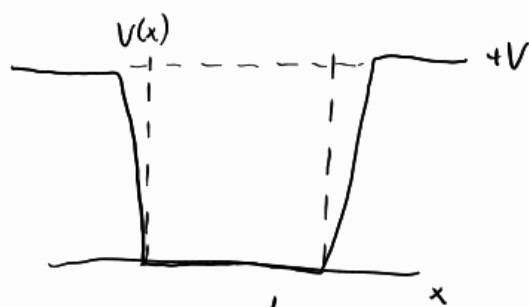
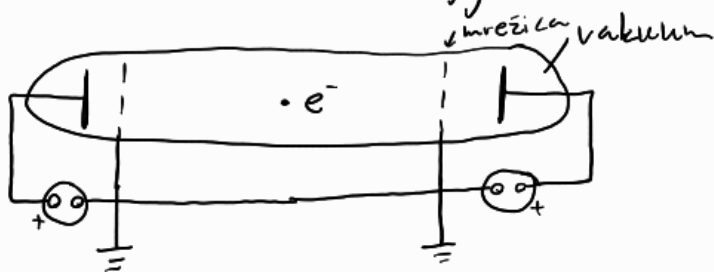


NE SKONČNA POTENCIALNA JAMA (LONEC)

(infinite well, particle in a box)

Elektrostatska analogija:



$$V(x) = \begin{cases} 0; & 0 \leq x \leq a \\ \infty; & \text{sicer} \end{cases}$$

- E delca samo kinetična ($V=0$)
- delec ne more bivati zunaj jame

rešujemo SSE:
$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E \psi(x)$$

$$\frac{d^2 \psi}{dx^2} = -\left(\frac{2mE}{\hbar^2}\right) \psi(x) \rightarrow \psi(x) = A \sin(kx) + B \cos(kx)$$

robna pogoji:

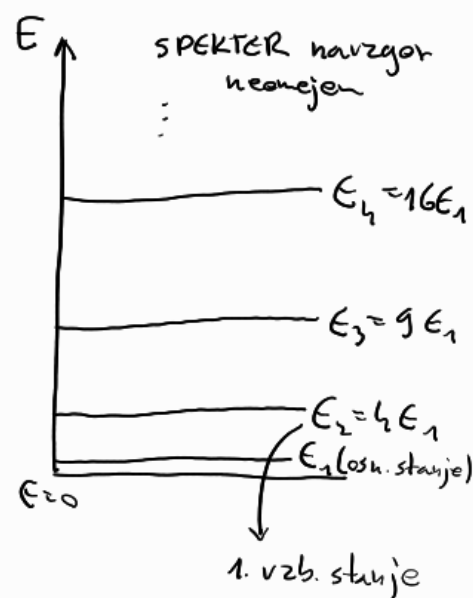
- $\psi(0) = 0 \Rightarrow B = 0$
- $\psi(a) = 0 \Rightarrow A \sin(ka) = 0 \Rightarrow ka = n\pi, n = 1, 2, 3, \dots$

↑
kvantno število

$$\Rightarrow E_n = \frac{p_n^2}{2m} = \frac{k_n^2 \hbar^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2m a^2} = n^2 E_1$$

$$E_n = n^2 E_1, E_1 = \frac{\pi^2 \hbar^2}{2m a^2}$$

E je kvantizirana!



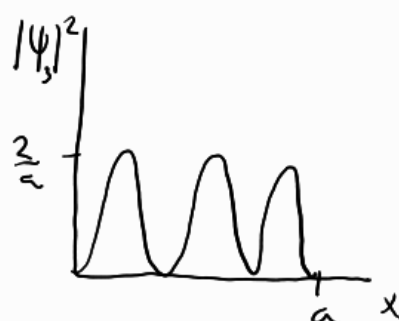
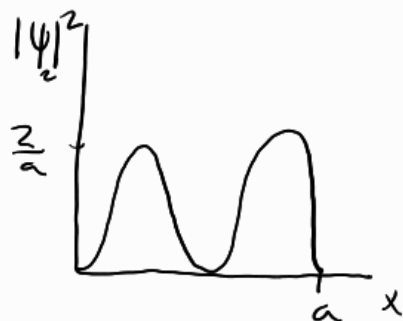
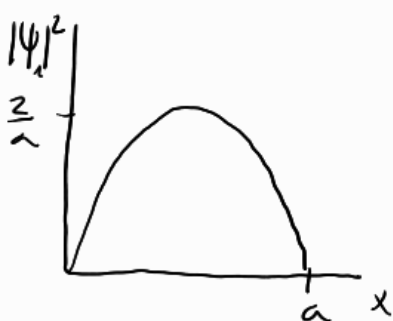
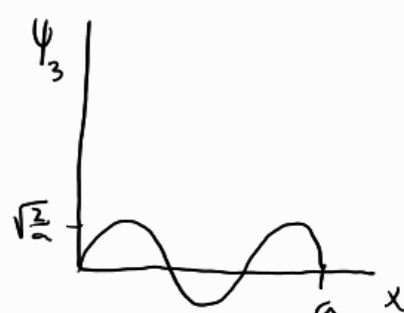
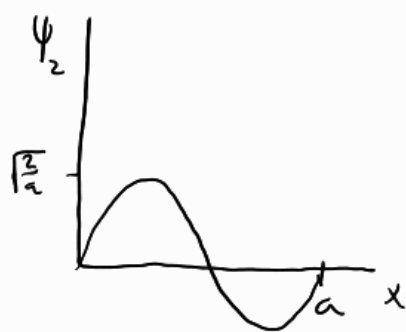
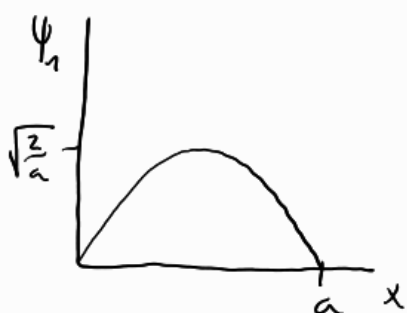
$$\psi_n(x) = A \sin\left(\frac{n\pi x}{a}\right), \quad n=1,2,\dots$$

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = \int_0^a \psi_n^2(x) dx = 1 \quad \forall n$$

recimo $A \in \mathbb{R}$

$$A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1 \rightarrow A = \sqrt{\frac{2}{a}} \rightarrow \boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)}$$

LASTNE FUNKC. ZA
 ∞ JAMO



Opombe:

- E osn. stanja $\neq 0$ (Heisenberg) $\rightarrow$ oceni: neutron $m = 1,67 \cdot 10^{-27} \text{ kg}$
3kratla (jedrski pot.) $a \approx 10^{-14} \text{ m}$

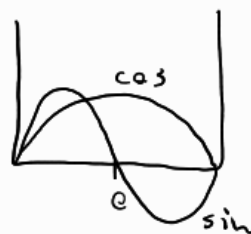
$$\int_0^a \psi_n(x) \psi_m^*(x) dx = \delta_{m,n} = \begin{cases} 1; & m=n \\ 0; & m \neq n \end{cases}$$

ORTOGONALNOST VF

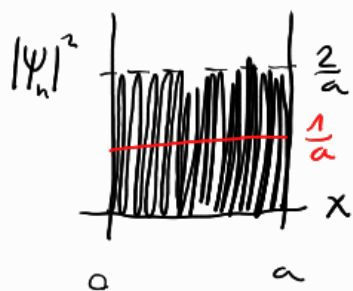
- namesto jame $[0, a]$ bi lahko vzeli jamo $[-a/2, a/2]$

$$\rightarrow \psi_n(x) \propto \begin{cases} \cos \frac{n\pi x}{a}; & n \text{ lih} \\ \sin \frac{n\pi x}{a}; & n \text{ sod} \end{cases}; \quad E_n = n^2 E_1$$

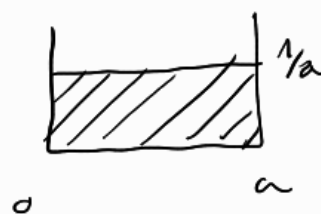
ostane enak



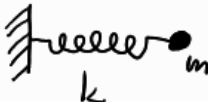
Korespondenčno načelo ($n \rightarrow \infty$)



$$\langle |\psi_n|^2 \rangle = \frac{2}{a} \langle \sin^2 \frac{n\pi x}{a} \rangle = \frac{1}{a} = f_{\text{klas.}}(x) \checkmark$$



LINEARNI HARMONIČNI OSCILATOR

Želimo kvantni analog 

- klasično: $V(x) = \frac{1}{2} kx^2$

$F(x) = -kx$ (Hooke)

$m\ddot{x} = -kx$ (Newton)

rešitev

$$x(t) = x_0 \cos(\omega_0 t + \delta)$$

$$\dot{x}(t) = -x_0 \omega_0 \sin(\omega_0 t + \delta)$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

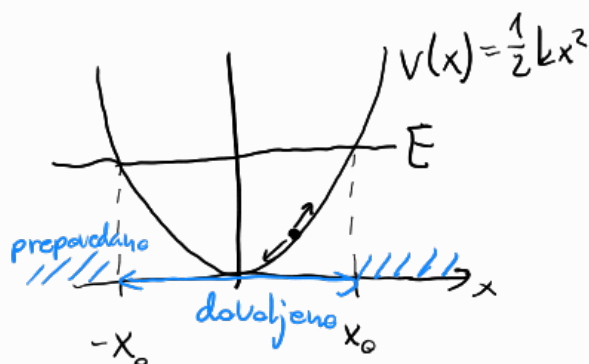
$$E = T + V = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} kx^2 =$$

$$= \frac{1}{2} m (\omega_0^2 \sin^2(\dots) + k \cos^2(\dots)) = \frac{1}{2} kx_0^2$$

$$\rightarrow V(x) = \sqrt{\frac{k}{m} (x_0^2 - x^2)}$$

klas. gibanje omejeno na $-x_0 \leq x \leq x_0$

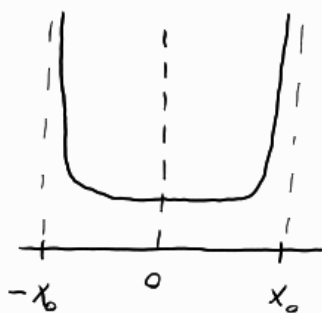
OBRAČALIŠČE



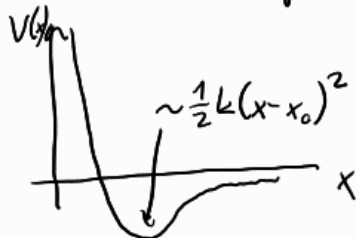
klas. verj. gostota

$$f_{\text{klas.}}(x) dx = \frac{dt}{T/2} = \frac{dx/\dot{x}}{T/2} = \frac{dx}{\sqrt{\frac{k}{m}(x_0^2 - x^2)} \pi \sqrt{\frac{m}{k}}} \rightarrow f_{\text{klas.}}(x) = \frac{1}{\pi \sqrt{x_0^2 - x^2}}$$

nihičias
 $T = \frac{2\pi}{\omega}$



LHO dober kot približek za realistične $V(x)$ -vaje



Kvantno:

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + \underbrace{\frac{1}{2} k x^2}_{V(x)} \psi = E \psi$$

hove spr:

$$\xi^2 = \frac{\sqrt{mk}}{\hbar} x^2$$

$$\lambda = \frac{2E}{\hbar} \sqrt{\frac{m}{k}} = \frac{2E}{\hbar \omega_0}$$

$$\rightarrow \frac{\hbar^2}{\sqrt{mk}} \frac{d^2 \psi}{d\xi^2} = \left(\frac{\sqrt{mk}}{\hbar} x^2 - \frac{2E}{\hbar} \sqrt{\frac{m}{k}} \right) \psi$$

$$\frac{d^2 \psi}{d\xi^2} = (\xi^2 - \lambda) \psi$$

predznak dobica značaj rešitve

$\xi^2 - \lambda < 0$: spremeni predznak ψ'' glede na ψ
 $\xi^2 - \lambda > 0$: ψ'' enako predznačen kot ψ

Ne glede na to: ψ definiran po vsej $\mathbb{R}$ (rešitev $\exists$ tudi v klas. prepovedanem obm.)

Ugibajmo: $\psi(\xi) = e^{-\xi^2/2}$

$$\psi'(\xi) = -\xi e^{-\xi^2/2}$$

$$\psi''(\xi) = -e^{-\xi^2/2} + \xi^2 e^{-\xi^2/2} = e^{-\xi^2/2} (\xi^2 - 1)$$

To izpolni enačbo, če $\lambda_0 = 1 \rightarrow 1 = \frac{2E_0}{\hbar \omega_0} \rightarrow E_0 = \frac{\hbar \omega_0}{2}$

$\uparrow$
 osn. stanje

$\hbar \omega_0$
 en. osn. st.

Podobno iščemo vzb. stanja:

$$\psi_n(\xi) = \text{polinom}(\xi) e^{-\xi^2/2}$$

(stopnje n)

$$\psi_n(\xi) = \left(\frac{m \omega_0}{\pi \hbar} \right)^{1/4} \sqrt{\frac{1}{2^n n!}} H_n(\xi) e^{-\xi^2/2}$$

normalizacijski faktor

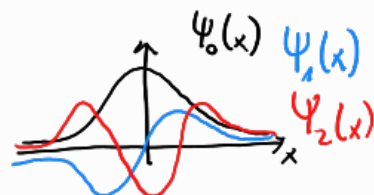
Hermitovi polinomi

$$H_0(\xi) = 1$$

$$H_1(\xi) = 2\xi$$

$$H_2(\xi) = 4\xi^2 - 2$$

$\vdots$



(tudi $-H_n(\xi)$ je ok)

$$E_n = \hbar \omega_0 \left(n + \frac{1}{2} \right)$$

$n = 0, 1, \dots$

